



POWER SERVICES ENGINEERING PRODUCT SERVICE

GE Power
TIL 1603-R2
12 FEBRUARY 2019

Compliance Category - **C**

Timing Code - **5**

TECHNICAL INFORMATION LETTER

R0 EROSION AND WATER INGESTION RECOMMENDATIONS

APPLICATION

All F-class gas turbines excluding 6F.01, 7F.04-200 and 7F.05.

PURPOSE

To advise operators of the inspection and maintenance considerations necessary to mitigate R0 erosion effects, and to inform them of a new online water wash (OnWW) system that reduces the rate of erosion in the principal area of concern. Failure to properly monitor or repair R0 erosion could result in cracking of the R0 blades and risks of compressor damage.

REASON FOR REVISION

The reason for this revision is to improve the clarity of the recommendations regarding online water wash frequency and mold inspections for specific F-class compressor configurations.

COMPLIANCE CATEGORY

M - Maintenance

Identifies maintenance guidelines or best practices for reliable equipment operation.

C - Compliance Required

Identifies the need for action to correct a condition that, if left uncorrected, may result in reduced equipment reliability or efficiency. Compliance may be required within a specific operating time.

A - Alert

Failure to comply with the TIL could result in equipment damage or facility damage. Compliance is mandated within a specific operating time.

S - Safety

Failure to comply with this TIL could result in personal injury. Compliance is mandated within a specific operating time.

TIMING CODE

1

Prior to Unit Startup / Prior to Continued Operation (forced outage condition)

2

At First Opportunity (next shutdown)

3

Prior to Operation of Affected System

4

At First Exposure of Component

5

At Scheduled Component Part Repair or Replacement.

6

Next Scheduled Outage

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BACKGROUND DISCUSSION

As the forward-most rotating blade in an F-class gas turbine compressor, the R0 (Rotor stage 0 blade) is the first to see airborne materials ingested through the inlet (see Figure 1). In addition to solid objects that may create immediate and substantial impact damage, ingested fluids and liquid droplets may also affect the blades, though more gradually, accumulating erosion primarily along the leading edge of the R0 (see Figure 2).

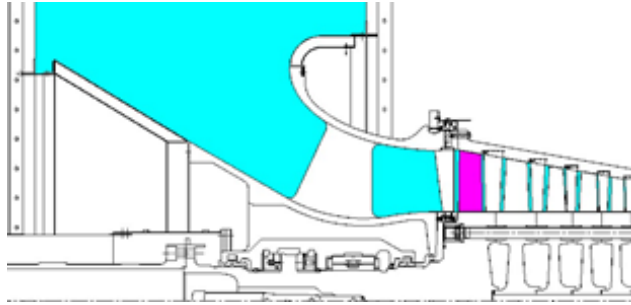


Figure 1: Schematic of Inlet and Forward Compressor

Common sources of water ingested through the gas turbine compressor inlet during operation include OnWW, fogging/SPRITS system output, and evaporative cooling system carry-over. Water ingestion could also result from unintended sources such as the leakage of rainwater through compromised inlet ducting seals or malfunctioning accessory equipment.

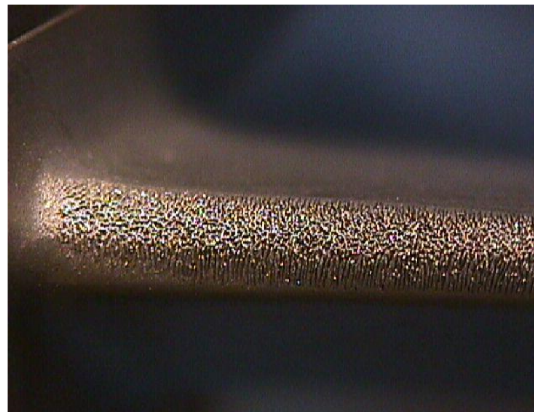


Figure 2: R0 Leading Edge Erosion

Leading edge erosion degrades the blade's material capability by creating a dense network of surface pits (see Figure 3) and has resulted in compressor event forced outages, with cracks initiating from erosion pits at the root of the R0 airfoil.

This TIL supersedes prior TILs 1303, 1389, 1400 and 1401, and provides consolidated recommendations regarding monitoring R0 erosion levels and repairing or mitigating such erosion. Failure to follow these recommendations could result in excessive erosion, which, in turn, could lead to subsequent cracking and possible liberation of R0 blades with potential downstream damage to the gas turbine.

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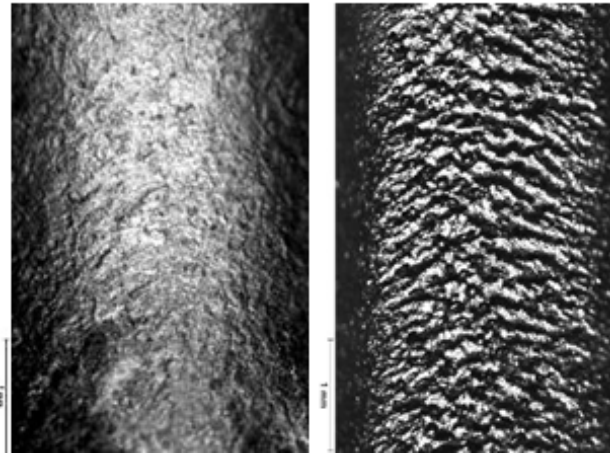


Figure 3: Erosion Mold Inspections (no erosion, 7 mils erosion)

RECOMMENDATIONS

Recommendations for monitoring erosion levels caused by known sources of potential water droplet introduction to the inlet air stream are given in the following section. The recommended method for erosion quantification is through mold impressions and GE Power Services provides a process for evaluating these molds. The accumulation of erosion may at some point require leading edge surface renewal prior to reaching a degradation limit. Typically this activity may be performed at Major Inspection intervals, but additional inspections at regular intervals may be necessary to confirm a modest degradation rate. Contact your GE Power Services provider for scheduling of this activity.

On-line Water Wash

Compressor fouling results from the depositing of airborne contaminants onto the compressor flow path surfaces and it affects a decline in aerodynamic efficiency and, thus, gas turbine performance. Water washing of the compressor is necessary to reduce the fouling effects and maintain a desired level of output. However, water washing performed on-line inherently injects water droplets into the compressor at a high relative velocity, thereby creating a source of R0 leading edge erosion.

TIL 1323 advises performing a modification to the original online water wash system to reduce the volume of water impinging on the R0 leading edge root location (the area susceptible to cracking from erosion). With this modification, erosion rates are improved, but still require special maintenance intervals. GE has now engineered and validated a new online water wash system that imparts significantly reduced erosion at the R0 root and requires no maintenance for up to 1000 cumulative hours of washing. This allows for daily water washing (15 minutes per 48 fired hours) with no R0 erosion maintenance necessary until the scheduled Major Inspection interval. The new configuration has been validated through accelerated field testing which confirmed the reduced rate of erosion while maintaining the same cleaning effectiveness as the baseline system. The system has been introduced into GE F-class production as well as retrofitted into field units as a system upgrade. Contact your GE Power Services representative for information on this upgrade.

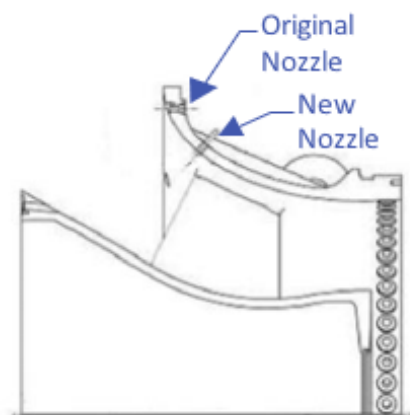


Figure 4: Relocated and Redesigned Spray Nozzles as Part of Upgraded OnWW System

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For current F-class units operating with the original OnWW system (which should be modified per TIL 1323), erosion levels should be inspected at regular intervals of OnWW by taking mold impressions of the R0 leading-edge root location. See “Combined Wet Inlet Operation” for inspection interval information. Contact your GE Power Services representative to arrange this inspection. Mold inspections should be followed by fluorescent penetrant inspection (FPI) to ensure against existing cracks.

OnWW should be performed based on desired output degradation limits as well as environmental effects to the forward compressor. When fouling deposits contain corrosive elements, washing can help minimize corrosive attack.

General guidelines for OnWW duration and frequency are as follows:

- 5 minutes per 48 Fired Hours, if standard R0's and the Gen-1 WW system are in place
- 15 minutes per 48 Fired Hours, if standard R0's and the Gen-2 WW system are in place (available through CM&U)
- 15 minutes per 48 Fired Hours, if enhanced R0s (ECP 2, 2+3, 4, 5) are installed with either the Gen-1 WW or the Gen-2 WW system

These durations are advised to balance erosion rates and output degradation. Shorter durations or longer intervals between washing may be scheduled at the operator's discretion based on minimal performance degradation or corrosion concerns. Conversely, longer or more frequent water washes could result in a greater rate of erosion, requiring more frequent repairs. For this reason, the upgraded OnWW system now available generally limits daily washes through control system permissives.

Inlet Fogging/SPRITS Operation

Water carry-over into the compressor may originate from inlet fogging systems. These systems introduce water droplets into the air stream that may coalesce and contribute to R0 leading edge erosion. Erosion rates will vary based on spray ratio, maintenance practices and operating frequency.

For units utilizing fogging systems, GE recommends that R0 blades be inspected with mold impressions and FPI at regular intervals of cumulative hours of fogger operation as defined in “Combined Wet Inlet Operation”.

Evaporative Cooler Operation

Though evaporative cooler systems are intended to cool the air and introduce moisture (not water droplets) for increased mass flow and efficiency, inadvertent water carry-over from this system may affect R0 erosion. GE recommends monitoring the evaporative coolers for proper operation and any evidence of water carry-over. This evidence may include water stains, puddles, or streaking in the filter inlet house or bell mouth area; deposits on filter or cycle deck material; or missing, damaged, or misaligned filter or cycle deck material.

Any evidence of water carry-over should be dispositioned and eliminated by GE or the manufacturing vendor. Refer to TILs 1285 and 1399 for additional information on evaporative cooler maintenance and annual commissioning.

R0 erosion molds and FPI should be performed between 100 and 300 hours of cumulative operation of the evap system. For units with identified water carry-over, the inspection should be repeated at this interval or as determined based on the mold impressions. If there is no significant erosion and it is continually verified that the evap system is generating no carry-over, no continual R0 mold inspection is necessary.

Combined Wet Inlet Operation

Many operators may use a combination of water-introducing systems. R0 erosion inspections should include both mold impressions and FPI and should be performed when cumulative wet inlet operation time reaches 100 hours as defined by the following formula:

Combined Wet Time = $4 \times (\text{OnWW hours}[A]) + .1 \times (\text{OnWW hours}[B]) + (\text{Fogger hours}[C]) + .2 \times (\text{SPRITS hours}[D]) + (\text{Evap hours}[E])$

- Where [A] pertains to the legacy OnWW system, [B] pertains to the upgraded OnWW system, [C] applies for non-GE foggers, [D] applies for GE fogging/SPRITS systems, and [E] applies only if carry-over has been present or the evaporative cooler system has not been commissioned or maintained per TILs 1285 and 1399.

From this, it can be seen that if an upgraded OnWW system is applied and evaporative coolers have been commissioned and maintained, then recurring mold inspections are primarily dependent on the accumulation of Fogging/SPRITS operation. These systems may have significant variability based on manufacturer, vintage, system tuning and modifications. As such, following initial inspections, parameters C and D may be adjusted per GE disposition (see Figure 5).

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These interval recommendations do not apply to unit operation with p-cut R0 blades. For these blades, erosion repairs can be performed at Major Inspection intervals or earlier as preferred.

	System	Usage Guidelines	Stand-Alone Mold Inspection Interval	Combined Operational Mold Inspection Interval
A	Original OLWW (Gen-1)	5 min / 48 FH	23-30 hrs OnWW or annually	Accumulated wet conditioning time: $4A + \frac{B}{10} + C + \frac{D_{[1]}}{5} + E_{[2]} = 100 \text{ hrs}$ [1]May be adjusted based on initial inspections [2]Not applicable once no carry-over verified
B	Upgraded OLWW (Gen-2)	15 min / 48 FH	1000 hrs OnWW or MI's	
C	Non-GE Fogging	N/A	100 hrs fogging [1]	
D	GE Fogging / SPRITS	Per GE Specs	500 hrs fogging [1]	
E	Evaporative Cooling	TIL1285, TIL1389	Initial commissioning: 100-300 hrs evaps Repeat if carry-over is observed	

Figure 5: Standard R0 Inspection Guidelines for Wet Inlet Conditioning

As accumulated wet conditioning time approaches the specified interval limit, the erosion inspection (FPI & molds) should be scheduled with your GE Services representative. The Service Representative will facilitate evaluation and disposition of the findings.

It should be anticipated that wet inlet operation will result in some level of R0 erosion for which the disposition could be a reduced interval for subsequent inspection or even discontinuance of further wet inlet operation until the R0s can be blend repaired by qualified GE Energy Services technicians. The intent is to avoid continued erosion beyond a threshold for cracking risk. Design analysis and field experience have defined this threshold at a measured erosion depth of .008 inches. Blade repairs performed by GE Services may be conducted in-situ; however, with a seed set of R0s, removal and repair in a GE Service Center will often minimize outage delays.

The leading edge R0 inspections and molds highlighted in this TIL are no longer recommended if the unit has enhanced R0 blades.

PLANNING INFORMATION

Compliance

- Compliance Category: **C**
- Timing Code: **5**

Manpower Skills

For R0 erosion molds, a GE Field Engineer or GE LES Inspector should perform the procedure per GE specifications.

R0 FPI should be performed by qualified inspectors from GE Services, LES or a qualified vendor.

For R0 erosion blending, the work should be performed by qualified GE Service Center blade technicians. Shot peening is also required post-blending, and a qualified vendor should be used for this operation.

For OnWW upgrade to Gen-2, the CM&U process should be followed for quoting of labor and parts. On-site machining will be necessary (and can be sourced by GE Power Services manager).

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Parts

A molding kit will be necessary to conduct the R0 erosion replications. Contact your GE Services representative to procure materials and resources for the inspection and to set up proper evaluation and disposition by GE Engineering.

Likewise, for OnWW upgrade to Gen-2, contact your GE Services representative to commence the quotation process.

Special Tooling

FPI chemicals should be provided by the contracted inspectors.

Erosion molding materials are provided in the referenced kit. See Parts.

Tooling for erosion blending supplied and utilized by GE Service Center blade technicians.

Reference Documents

TILs 1285, 1323, and 1399

Scope of Work

R0 erosion molds and FPI can be performed concurrently with TIL 1509 inspections.

- R0 erosion molds typically take at most ½ shift or 6 man-hours to complete.
- FPI of R0s typically requires one 12-hour shift.

Leading edge blending for R0 erosion can be done at site or with the blades sent to a GE Service Center. Typical duration for blending in the unit is four days and requires, at a minimum, that the inlet bellmouth be jacked approximately two feet. Therefore, if performed outside of an MI, a spare set of R0s may be preferred. Alternatively, the blending may be performed in a GE Service Center in as little as two days.

TIL DISPOSITION

Disposition of TILs should be entered in local records and also in GE Power ServiceNow. Follow the below instructions for entering the disposition record;

- Log into the Power ServiceNow at https://gepowerpac.service-now.com/til_new/ using your GE SSO number and password.
- Select "TIL Disposition".
- Click on the TIL for the serial number you want to update.
- Choose the most appropriate "Disposition Status" and enter "Disposition Notes".
- Click "Save".

Contact your local GE Services representative for assistance or for additional information

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